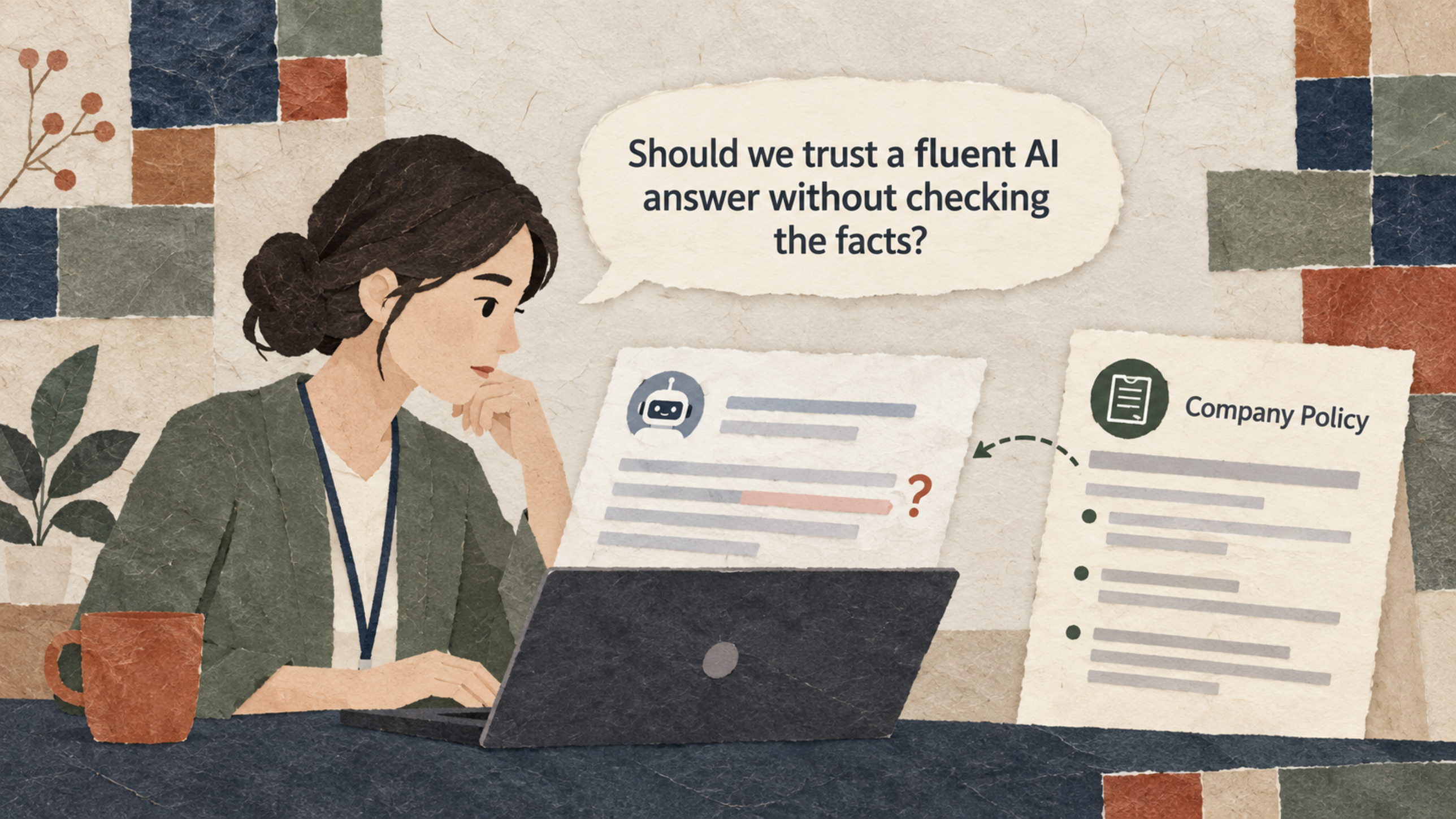

Should we trust a fluent AI answer without checking the facts?
?
Company Policy

What problem are we solving?

3 Hours (180 minutes) - Hour 1 (50 min): Comprehensive Concept Lecture & Live Demonstration - Hour 2 (50 min): Guided Hands-On Lab (Basic Exercise: Customer Review Comparison) - Hour 3 (50 min): Applied Business Challenge & Peer Debrief (Advanced Exercise: Multi-Issue Escalation)

SITUATION

The problem

Business staff need useful AI-written communication, but fluent output can contain unsupported facts. They must know when to search, when to draft with AI, and what a person must verify.

TARGET

Learning goal

Learn what ChatGPT can and cannot do, understand how it generates text, and build safety habits for daily office tasks.

Practice material: Use the practice material supplied in the session Markdown.

Use these supplied facts

Do not replace missing information with guesses.

CASE MATERIAL

Public Data Source

- PyCaret Open Amazon Customer Reviews Dataset (CSV) (https://raw.githubusercontent.com/pycaret/pycaret/master/datasets/amazon.csv) - Real open customer reviews and ratings.

BOUNDARY

Use the supplied facts

Business staff need useful AI-written communication, but fluent output can contain unsupported facts. They must know when to search, when to draft with AI, and what a person must verify.

What students need to understand

Key explanations taken from the active session lesson.

CORE IDEA

How LLMs Actually Work (15 min)

- The Predictive Autocomplete Analogy: Most business professionals mistakenly believe ChatGPT searches an encyclopedia database or "thinks" like a human. In reality, Large Language Models predict the next most likely word token based on statistical patterns learned from vast training datasets. When an LLM generates text, it calculates probability distributions across billions of parameters. Because it generates text probabilistically rather than retrieving hard records, it produces fluent language, but has no inherent sense of truth.

CORE IDEA

ChatGPT vs. Google Search - The Strategic Distinction (15 min)

- Decision Framework: Teach students when to search vs. when to prompt.
- Use Traditional Search (Google): For verifiable historical facts, live stock quotes, statutory regulations, telephone numbers, and cited scientific research.
- Use Generative AI (ChatGPT): For drafting executive correspondence, translating tone (e.g., technical jargon to layperson English), summarizing unformatted text, and brainstorming strategic angles.
- The Modern Business Hybrid Workflow: (1) Gather verified facts from primary company sources -> (2) Feed those verified facts into ChatGPT to draft and format the communication.

What students need to understand

Key explanations taken from the active session lesson.

CORE IDEA

The Hallucination Problem & Enterprise Risk (10 min)

- What is a Hallucination? Plausibly sounding, highly confident statements generated by AI that are completely false. Case Study - Air Canada Lawsuit (2024): An Air Canada customer service chatbot invented a non-existent bereavement refund policy. A customer relied on the chatbot's advice and sued when the refund was refused. The Canadian tribunal ruled Air Canada legally liable for its chatbot's hallucination. Managerial Golden Rule: AI is a brilliant junior copywriter, but a terrible factual auditor. Never approve an AI-generated document without personally verifying dates, names, and calculations.

CORE IDEA

Instructor Live Demonstration Script & Discussion (10 min)

- Step 1 (Naive Request): Type "Write an apology email to an angry customer whose delivery was late." Note the empty platitudes ("We value your business") and lack of actionable next steps. Step 2 (Engineered Prompt): Type a structured prompt specifying Role ("Customer Support Lead"), Context ("4-day courier delay on subscription coffee"), Task ("Draft 3-paragraph email"), Constraints ("Acknowledge delay in sentence 1; offer $10 credit code [COFFEE10]; explain backup carrier policy in 2 bullets; under 150 words"). Discussion Question: "Why does setting a specific persona (like 'Senior Operations Lead') improve AI output quality?" (Answer: It narrows the model's vocabulary and probability distribution toward professional domain terminology).

What makes a prompt weak?

These examples come from the current session scenario.

AVOID

Weak example 1

```
Write an apology email to an angry customer whose
delivery was late.
```

A clearer, safer prompt

Keep the source, task, constraints, and output check visible.

USE

Improved example 1

```
You are the Customer Support Lead for an online
artisanal coffee subscription brand.
    A long-time subscriber experienced a 4-day
courier delay due to regional weather.

    Write a 3-paragraph apology email following
these rules:
    1. Acknowledge and apologize for the shipping
delay in the very first sentence.
    2. Offer a $10 store credit code [COFFEE10]
toward their next monthly renewal.
    3. Explain our backup carrier routing policy
in 2 concise bullet points.
    4. Maintain an empathetic, professional tone
and keep the total length under 150 words.
```

Exercise 1: Follow the steps

Use the complete case and instructions from the session Markdown.

GUIDED PRACTICE

Exercise 1: Follow the steps

- Scenario: Using the PyCaret Amazon Customer Reviews dataset from this session, select one 5-star positive review and one 1-star negative complaint.
- Task: Test a naive prompt versus the engineered Customer Support Lead prompt provided in the lesson.
- Deliverable: A Prompt Comparison Scorecard table evaluating both outputs across usefulness, accuracy, and executive tone (1-to-5 scale), accompanied by a 1-sentence diagnostic note.

Exercise 2: Try a small variation

Use the complete case and instructions from the session Markdown.

APPLIED PRACTICE

Exercise 2: Try a small variation

- Scenario: From the same Amazon Customer Reviews dataset, identify a complex negative review that cites multiple simultaneous operational failures (e.g., damaged product packaging, a delayed shipment, and an unresponsive delivery courier).
- Task: Engineer an advanced prompt instructing ChatGPT to act as a Senior Customer Operations Lead. Apply strict negative constraints (no excuses, no blaming shipping couriers), propose a tiered resolution (immediate product replacement plus a promotional discount code), and generate a simultaneous 3-bullet internal operational alert for the warehouse fulfillment team.
- Deliverable: A two-part deliverable: (1) A polished, empathetic customer reply under 150 words, and (2) A 3-bullet internal operational escalation note.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W1_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- Introduction: Explain why a generic AI reply to a customer review can hurt the business, and what you want a better prompt to do.
- Method: Copy the basic prompt, the Customer Support Lead prompt (Exercise 1), and your Senior Customer Operations Lead prompt (Exercise 2).
- Results: Include a scorecard table that rates both Exercise 1 outputs from 1 to 5 on usefulness, accuracy, and executive tone, with a one-sentence diagnostic note. Add the Exercise 2 customer reply (under 150 words) and the 3-bullet internal alert.
- Conclusion: Name the prompt element that improved the reply most, and one part you still had to check or fix yourself.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- See how giving clear background information produces much more relevant, professional, and practical answers.